

If I Had to Master SQL Joins in 1 Day Before an Interview

What are SQL JOINS?

JOINS combine rows from two or more tables based on related columns (keys). Think of tables as separate spreadsheets - JOINS merge them horizontally where values match, letting you analyze relationships like customers→orders→products.

Why JOINS matter: 70-80% of SQL interviews test JOINS because real analytics requires combining multiple tables.

🎓 JOIN CONCEPTS - Core Principles

1. Table Relationships

```
customers  orders
id | name  id | cust_id | amount
1  | Alice 101| 1       | 100
2  | Bob   102| 1       | 200
   |       103| 2      | 150
```

Foreign Key: `orders.cust_id` references `customers.id`

2. JOIN Logic

- **Matching:** Find rows where `table1.key = table2.foreign_key`
- **Preservation:** LEFT keeps all left table rows, INNER keeps only matches
- **NULL Handling:** Non-matching rows get NULL values

3. Visual Understanding

```
customers ●———● orders  INNER = ● (matches only)
customers ○———● orders  LEFT  = ○● (all customers)
customers ●———○ orders  RIGHT = ●○ (all orders)
```

📊 Complete Schema + Sample Data (SQLFiddle Ready)

```
CREATE TABLE customers (id INT, name VARCHAR(50), city VARCHAR(50), segment VARCHAR(20));
INSERT INTO customers VALUES
```

```
(1,'Alice','NY','VIP'), (2,'Bob','LA','Regular'),
(3,'Carol','NY','VIP'), (4,'David','SF','Regular'), (5,'Eve','LA','Trial');
```

```
CREATE TABLE orders (id INT, cust_id INT, amount DECIMAL(10,2), order_date DATE);
```

```
INSERT INTO orders VALUES
```

```
(101,1,100.00,'2026-01-01'), (102,1,200.00,'2026-01-02'),
(103,2,150.00,'2026-01-03'), (104,3,50.00,'2026-01-04');
```

6 JOIN TYPES - Concepts + Syntax + Examples

JOIN	CONCEPT	SYNTAX	RESULT	BUSINESS USE
INNER	Only matching rows	<code>c INNER JOIN o ON c.id=o.cust_id</code>	Alice(2), Bob(1)	Active customers
LEFT	All left + matches	<code>c LEFT JOIN o ON c.id=o.cust_id</code>	Alice,Bob,Carol,David,Eve	LTV/Churn analysis
RIGHT	All right + matches	<code>o RIGHT JOIN c ON c.id=o.cust_id</code>	Same as LEFT (swapped)	Rare
FULL	Everything + NULLs	<code>c FULL JOIN o ON c.id=o.cust_id</code>	All customers + all orders	Reconciliation
CROSS	Every combination	<code>c CROSS JOIN o</code>	5×4=20 rows	Rare
SELF	Same table	<code>e JOIN m ON e.mgr_id=m.id</code>	Hierarchies	Org charts

JOIN CONCEPT DEEP DIVE

1. How JOINS Work (Step-by-Step)

Step 1: Cartesian Product (all combinations)

Step 2: Filter WHERE condition matches (`c.id = o.cust_id`)

Step 3: Return matching rows (INNER) OR preserve rows (LEFT)

Example Trace:

Alice × [101,102,103,104] → matches 101,102 → Alice/100, Alice/200

Bob × [101,102,103,104] → matches 103 → Bob/150

Carol × [101,102,103,104] → no match → NULL (LEFT JOIN)

2. Key Decisions

"Revenue by city?" → ALL cities? → LEFT JOIN
Only cities WITH revenue? → INNER JOIN

3. Aggregation Rules

LEFT JOIN + SUM/COUNT → GROUP BY left table columns
COUNT(o.id) → NULLs become 0
SUM(o.amount) → NULLs crash → COALESCE(SUM(o.amount),0)

! TOP 5 JOIN MISTAKES (Interview Killers)

- ✗ 1. SELECT * FROM c JOIN o → Too many columns
✓ SELECT c.name, SUM(o.amount) FROM c JOIN o
- ✗ 2. SUM(amount) without GROUP BY → Error!
✓ GROUP BY c.id, c.name
- ✗ 3. LEFT JOIN then WHERE o.amount > 0 → Becomes INNER!
✓ WHERE COALESCE(o.amount,0) > 0
- ✗ 4. No aliases → c.id vs o.id ambiguity
✓ customers c, orders o
- ✗ 5. Forgetting COALESCE → LTV shows NULL
✓ COALESCE(SUM(o.amount),0) as ltv

🕒 12-HOUR MASTERY PLAN

PHASE 1: CONCEPTS + SYNTAX (6-9 AM)

Hour 1: Schema + 6 JOINS x5 each (60 min)

- ✓ INNER, LEFT, RIGHT, FULL, CROSS, SELF
- ✓ Screenshot each result

Hour 2: Error Patterns (60 min)

- ✓ Fix 15 broken queries above
- ✓ Memorize: "LEFT+WHERE NULL = anti-join"

Hour 3: Verbal Practice (60 min)

✓ Explain each JOIN x3: "LEFT preserves..."

PHASE 2: PATTERN MASTERY (9AM-3PM)

40 Basic: SQLZoo Joins tutorial

60 Medium: DataLemur + LeetCode Easy

30 Advanced: StrataScratch + LeetCode Medium

PHASE 3: MOCK INTERVIEWS (3-8PM)

5×10min screens:

1. "Revenue per city (incl zeros)"
2. "Inactive customers"
3. "Top customers per segment"
4. "New vs returning"
5. "90-day churn"

12 KEY INTERVIEW TEMPLATES

Question	Query
Active revenue	<code>INNER JOIN orders GROUP BY customer</code>
LTV by city	<code>LEFT JOIN GROUP BY city COALESCE(SUM(amount),0)</code>
Churn list	<code>LEFT JOIN WHERE orders.id IS NULL</code>
New customers	<code>LEFT JOIN prev_month WHERE prev.id IS NULL</code>
Top 3/city	<code>ROW_NUMBER() OVER(PARTITION BY city)<=3</code>

PERFECT INTERVIEW RESPONSE (30 seconds)

1. "Two tables: customers(id,name,city) + orders(cust_id,amount)"
2. "Need revenue PER CITY including zero-revenue cities"
3. "LEFT JOIN orders preserves all customers → GROUP BY city"
[WRITES: `c LEFT JOIN o ON c.id=o.cust_id GROUP BY c.city`]
4. "NY=350, SF=0 confirms zeros included"
5. "Gives complete regional performance"

✓ SUCCESS CHECKLIST

- ☐ Understand 6 JOIN types + when to use
- ☐ Fix all 5 common errors from memory
- ☐ Solve 150 problems (90% accuracy)
- ☐ Explain any query in <30 seconds
- ☐ Pass 5 mock screens under 10 mins